

Iulian Oleniuc

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Education

Master's Degree

2023–2025

Institution: Faculty of Computer Science Iași

Program: Advanced Studies in Computer Science

Thesis: In my [MSc thesis](#), “A Hybrid Key-Policy Attribute-Based Encryption Scheme for Universal Circuits, Based on Monotone Span Programs,” building on top of the Hu–Gao and Goyal et al. schemes, I developed the **pairing-based KP-ABE** scheme with, I argue, **the highest degree of generalization**. A paper on part of this work was accepted at the **CSJM 2025** journal.

Thesis grade: 9.5

Bachelor's Degree

2020–2023

Institution: Faculty of Computer Science Iași

Thesis: In my [BSc thesis](#), “Heuristic Optimizations for Boolean Formulas with Applications in Attribute-Based Encryption,” I developed four heuristics to **optimize monotone Boolean formulas**. These optimizations enhanced the performance of a **circuit-based KP-ABE** scheme by up to 60% in real-world scenarios. A paper on this work was accepted at the **KES 2023** conference.

Thesis grade: 10

High School

2016–2020

Institution: Colegiul Național “Emil Racoviță” Iași

Specialization: Mathematics–Informatics

Experience

Status 32

Oct 2025–Present

Designed and implemented end-to-end features for a **Customer Relationship Management** system, including modules such as an office scheduler and a train route registry.

Built responsive frontend interfaces using **React**, leveraging **DevExtreme** components (data grids, forms), along with **Zustand** for state management and **TanStack Query** for data fetching and caching.

Developed RESTful CRUD APIs with **.NET**, while working with **Microsoft SQL Server Management Studio** for basic database management.

Collaborated in sprint-based development and gained exposure to **Azure DevOps** and **CI/CD** pipelines.

Languages

- Romanian — Native
- English — Advanced

Skills

- Problem solving
- Teaching
- Technical writing
- Teamwork
- Critical thinking

Knowledge

- Algorithms
- Data structures
- Cryptography
- Machine learning
- Neural networks
- Web developing
- Relational databases
- Object-oriented programming

Technologies

- C++, Java, Python
- HTML, CSS, Tailwind
- JavaScript, TypeScript
- React, Vue
- Gatsby, Nuxt, Astro
- Firebase, Supabase
- PostgreSQL

Passions

- Cycling
- Books
- Hip-hop

Contests

Algorithms	High School–College
National Olympiad in Informatics: Participation (9th grade), Bronze (10th grade), Silver (11th grade), Qualification (12th grade)	
ICPC Regional Contest Iași: 1st place (3rd year of BSc)	

Publications

LSSS Limitations Over Boolean Circuits	2025
A paper co-authored with Alex Ioniță, published in CSJM, 33(1), 2025 . It offers a gentle introduction into the realm of monotone span programs, linear secret sharing schemes, and attribute-based encryption.	
Boolean Circuit Optimizations for ABE	2023
A paper co-authored with Alex Ioniță and Denis Banu, published at the KES 2023 conference. It introduces four heuristics for optimizing Boolean circuits in key-policy attribute-based encryption schemes.	

Projects

InfoGenius — Computer Science Blog	2017–Present
I founded InfoGenius in the 9th grade and have since written posts on topics such as famous algorithms, data structures, C++ programming, and National Olympiad problems, among others. With over 300 comments , the website has made a meaningful impact in helping people learn computer science, particularly in competitive programming.	
InfoGym — Teaching Computer Science	2022–2025
InfoGym is a programming circle founded by eMAG as part of the “ Hai la Olimpiadă! ” program, where I have delivered several lectures to middle and high school students. This involved creating instructional papers, including “ Introduction to Combinatorics ” and “ Trie & the Knuth–Morris–Pratt Algorithm ,” which are among my proudest works.	
FIICode — Organizing a Programming Contest	2021–2024
FIICode is a programming contest organized by students of the Faculty of Computer Science Iași. I contributed to the organizing and scientific committees for the 2021–2024 editions, focusing on algorithms. My responsibilities included designing problem ideas, generating test data, and preparing problem statements and editorials.	
CruceaRosielasi.ro — Website for the Iași Red Cross	2022
I developed a fast, user-friendly website for the Iași Red Cross using modern web technologies like Vue, Nuxt, and Firebase. The site includes an admin panel with a minimalist yet powerful content management system .	